SHORT COMMUNICATION

Day- and Night-time Noise Levels at a Commercial-Area Monitoring Station: Limit-Exceedance Classification and Regression with Classical Machine-Learning Models

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Abstract. Traffic noise above prescribed limits harms sleep and cardiovascular health, and India sets separate day (65 dB(A)) and night (55 dB(A)) limits for commercial areas. This exploratory study asks whether classical machine-learning models can (i) explain monthly day- and night-time equivalent noise level (L_{eq}), (ii) classify whether a month exceeds its limit, and (iii) forecast next month's level. We use one open CPCB/KSPCB file of twelve monthly summaries (April 2019–March 2020; 12 months × day/night = 24 observations), screen it for quality problems, and compare linear, ridge, decision-tree and k-NN regressors and logistic-regression, tree, k-NN and naive-Bayes classifiers with trivial baselines under leave-one-month-out and rolling-origin evaluation. The limit was exceeded in 11 of 12 night-time months and 7 of 12 day-time months. Night-time L_{eq} was only 1.7 dB(A) below day-time (Wilcoxon p = 0.13) although its limit is 10 dB(A) lower. No learned model beat the period-mean baseline when explaining L_{eq} (baseline RMSE 8.53; ridge 9.09, R^2 = -0.33), and the majority rule (F1 = 0.857) matched or beat every classifier (logistic regression F1 = 0.821, AUC = 0.52). In a six-month rolling forecast, ridge regression had the lowest RMSE (6.46 versus 7.91 for persistence) but the ranking changed with the penalty strength and rests on 12 predictions. Twelve monthly values are too few for machine learning to add value; the robust findings are persistent non-compliance and data-quality anomalies.

Keywords: noise pollution; equivalent noise level; CPCB limits; regression; logistic regression; k-NN; limit exceedance; time-series validation; SDG 3; SDG 11

1 Introduction

Environmental noise is a recognised health risk: reviews link long-term exposure, especially at night, with sleep disturbance, annoyance and cardiovascular disease [3, 4], and the WHO gives separate guideline values for road-traffic noise [2]. In India the Noise Pollution (Regulation and Control) Rules, 2000 set ambient limits as A-weighted equivalent levels; for commercial areas these are 65 dB(A) by day (06:00–22:00) and 55 dB(A) by night (22:00–06:00) [1]. Monthly summaries from continuous noise monitors are published on the open-government-data portal [5] but are rarely modelled.

We use one commercial-area station to ask three questions. **Q1**: how do day and night L_{eq} depend on the period of day, the month and the festival season? **Q2**: can a month's exceedance of the prescribed limit be classified from the same information? **Q3**: can last month's level forecast next month's? We use the classical models of an undergraduate course and always compare them with trivial baselines, because with so few observations a benefit must be demonstrated rather than assumed. Our contributions are: (i) a cleaned, quality-screened dataset; (ii) two leakage-aware evaluation protocols; (iii) an explicit, mostly negative, result; and (iv) a statement of what these data cannot answer (weekday effects). Section 2 reviews related work, Section 3 the data, Sections 4–5 the methods, Section 6 the results and Section 7 the discussion.

2 Related Work

Basner et al. [3] and Münzel et al. [4] review the auditory and non-auditory health effects of environmental noise, which motivates night-time limits. For time-ordered data, Bergmeir and Benítez [6] discuss when ordinary cross-validation is valid; we therefore pair a leave-one-month-out analysis (to explain levels) with forward-chaining evaluation (to forecast). The national COVID-19 lockdown began on 25 March 2020, in the last week of our data, so its effect cannot be separated here. All models were implemented with scikit-learn [7]; the algorithms follow [8].

3 Dataset and Preprocessing

3.1 Source and description

The data are monthly noise summaries from a CPCB/KSPCB commercial-area station on the Open Government Data (OGD) Platform India [5], supplied as *noise_data_csv.csv* (April 2019–March 2020). Each row is a month with, for day and night separately, the limit, L_{eq}, minimum (L_{min}) and maximum (L_{max}) levels in dB(A), a "% Increase" over the limit and the number of days (Table 1); day is 06:00–22:00 and night 22:00–06:00. After removing the header and footnote rows we obtained 12 months × 2 periods = 24 observations with no missing values.

Table 1. Raw variables in the CSV file (n = 12 months, each with a day and a night value).

Variable	Unit	Missing	Use in this study
Month	month	0 %	split, season (sin/cos)
Limit (day / night)	dB(A)	0 %	exceedance label (65 / 55)
L_{eq} day, night	dB(A)	0 %	regression target; label source
L_{min}, L_{max} (day, night)	dB(A)	0 %	quality screening only
% Increase	%	0 %	not used (derived from target: leakage)
No. of days	days	0 %	not used

3.2 Data quality

The published "% Increase" column agrees with the rule L_{eq} > limit in all 24 of 24 observations, so labels were derived from L_{eq} and the limit directly. Two anomalies were found. First, 9 of the 12 day-time rows (April–December 2019) report an identical L_{min} of 21.6 dB(A), a physically implausible constant for a commercial area that suggests a stuck or default reading; one of them (July 2019) also has L_{max} equal to L_{eq} (78.5 dB(A)). Second, the series is erratic: the mean absolute month-to-month change in L_{eq} is 10.9 dB (day) and 7.7 dB (night). We cannot tell whether this reflects real activity or measurement problems. Flagged rows were kept (the target itself shows no direct error) and tested in a sensitivity analysis (Section 6.5).

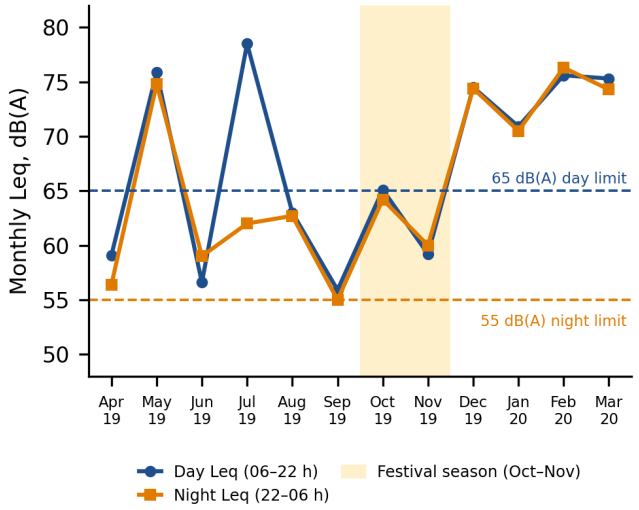


Fig. 1. Monthly day and night L_{eq}, April 2019–March 2020, with the 65 and 55 dB(A) limits (dashed). Yellow: festival season (October–November).

3.3 Labels and features

The label is c = 1 if L_{eq} > the limit of its period and 0 otherwise (night L_{eq} of exactly 55.0 dB(A) in September 2019 is "within limit"). Overall 18 of 24 observations (75%) exceed the limit (Fig. 2). Features (Table 2) are the period (night = 1), the month as sin/cos, a festival-season flag (October–November, Dasara–Deepavali) and, for forecasting, the previous month's value. Features were standardised inside each pipeline using training data only.

Weekday effects could not be modelled. The task mentions weekday analysis, but the file contains monthly aggregates only. We state this limitation rather than invent a proxy; daily or hourly records from the CPCB noise monitoring network would be needed.

Table 2. Engineered features; A = leave-one-month-out protocol, B = forecasting protocol.

Feature	Definition	Used in
night	1 if night-time row, else 0	A, B
month sin, cos	sin / cos(2π·month/12)	A, B
festival	1 for October and November	A, B
lag	previous month's L_{eq} of the same period (relative to the limit for classifiers)	B

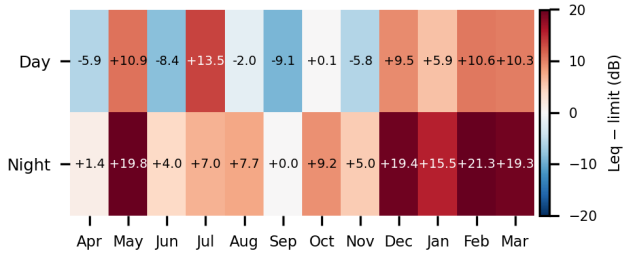


Fig. 2. Margin L_{eq} − limit (dB) by month and period. Red = exceedance, blue = within limit.

4 Methodology

Fig. 3 summarises the workflow. The equivalent level is the energy average of the A-weighted sound pressure p over a period T (p_0 = 20 µPa):

$$L_{eq} = 10 \log_{10} [\ (1/T) \int (p(t)/p_0)^2 \, dt\ ], \; c = 1 \text{ if } L_{eq} > L_{lim} \qquad (1)$$

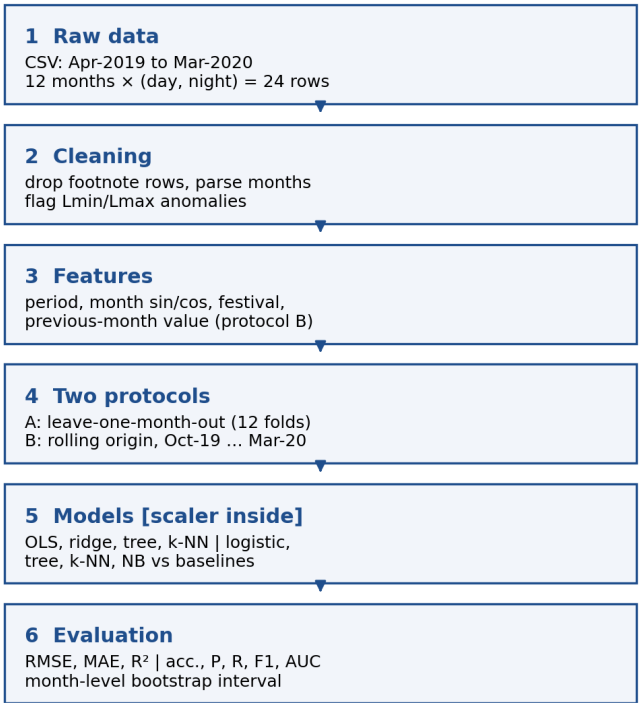


Fig. 3. Methodology pipeline (square brackets: scaler fitted inside the training folds).

4.1 Regression models

Multiple linear regression (OLS) fits $\hat{y} = \mathbf{w}^T\mathbf{x} + b$ by least squares. **Ridge** adds an L2 penalty against collinearity:

$$J(\mathbf{w}) = \Sigma_i (y_i - \hat{y}_i)^2 + \lambda \Sigma_j w_j^2 \qquad (2)$$

A **regression tree** (depth 3, at least 2 samples per leaf) splits the features to minimise within-node squared error; **k-NN regression** (k = 3, distance-weighted, Euclidean distance) averages the targets of the nearest training months. **Baselines:** the training mean per period and, for forecasting, *persistence* (next month = this month).

4.2 Classification models

Logistic regression models $P(c = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^T\mathbf{x} + b)$ with $\sigma(z) = 1/(1 + e^{-z})$ and an L2 penalty (C = 1). We also use a depth-3 **decision tree**, **k-NN** (k = 3) and **Gaussian naive Bayes**. Baselines are the majority class per period, persistence of last month's label, and "regression-then-threshold" (OLS predicts L_{eq}; flag a month if it exceeds the limit).

4.3 Evaluation metrics

Regression: MAE, RMSE and $R^2 = 1 - \Sigma(y - \hat{y})^2 / \Sigma(y - \bar{y})^2$. R^2 is negative whenever a model predicts worse than the mean of the evaluation data, which is common with few, noisy observations. Classification: accuracy, precision, recall, F1 and ROC-AUC; recall and F1 are emphasised because a missed exceedance is costlier than a false alarm. Uncertainty is shown by a month-level bootstrap (5,000 resamples).

5 Experimental Setup

Protocol A (explanatory): leave-one-month-out. Each of the 12 months is held out in turn (day and night rows together) and the model is trained on the other 11, giving 24 out-of-fold predictions. This is interpolation, not forecasting. **Protocol B (forecasting): rolling origin.** For each month t from October 2019 to March 2020, models are trained only on earlier months that have a lag value (5 to 10 months) and predict both periods of month t (12 predictions). **Hyper-parameters** were fixed in advance (ridge λ = 1; tree depth 3; k = 3; C = 1) because 12 months are far too few for nested tuning; Section 6.5 shows their sensitivity. Preprocessing sits inside a scikit-learn Pipeline. **Environment:** Python 3.12, scikit-learn, pandas, NumPy, SciPy, Matplotlib; random seed 42. The accompanying code regenerates every number and figure.

6 Results

6.1 Descriptive findings

Table 3. Descriptive statistics of monthly L_{eq} (dB(A)); margin = L_{eq} − limit; |Δ| m-o-m = mean absolute month-to-month change.

Period	Mean	Min–max	Over limit	Mean margin	|Δ| m-o-m
Day	67.5	55.9–78.5	7/12	+2.5	10.9
Night	65.8	55.0–76.3	11/12	+10.8	7.7

The night-time limit was exceeded in 11 of 12 months and the day-time limit in 7 of 12 (Table 3, Figs 1–2); only September 2019 had no exceedance in either period. Night L_{eq} was 1.7 dB(A) below day L_{eq} on average, a difference that is not significant (Wilcoxon signed-rank $p = 0.13$); because the night limit is 10 dB(A) lower, the night margin (+10.8 dB) is larger than the day margin (+2.5 dB). Festival months (October–November) were not noisier in monthly averages: the exceedance rate was 75% in both festival and other months and the mean margin was +2.1 versus +7.5 dB, although L_{max} reached 104.4 dB(A) (day) in October 2019. Monthly L_{eq} averages dilute such short events.

6.2 Explaining Leq (Protocol A)

Table 4. Leave-one-month-out regression of L_{eq} (dB(A); 24 out-of-fold predictions). Best RMSE in bold.

Model	MAE	RMSE	R^2
Period mean	**7.93**	**8.53**	**-0.18**
OLS	7.83	9.26	-0.39
Ridge	7.68	9.09	-0.33
Decision tree	10.93	12.35	-1.46
k-NN	8.30	9.69	-0.52

No learned model beat the period-mean baseline (RMSE 8.53): ridge reached 9.09 (R^2 -0.33), OLS 9.26, k-NN 9.69 and the tree 12.35 (Table 4, Fig. 4). All R^2 values are negative: out of sample, month and festival information predicts worse than the period mean. The bootstrap 95% intervals of ridge (6.56–11.28) and the baseline (7.01–9.74) overlap almost completely.

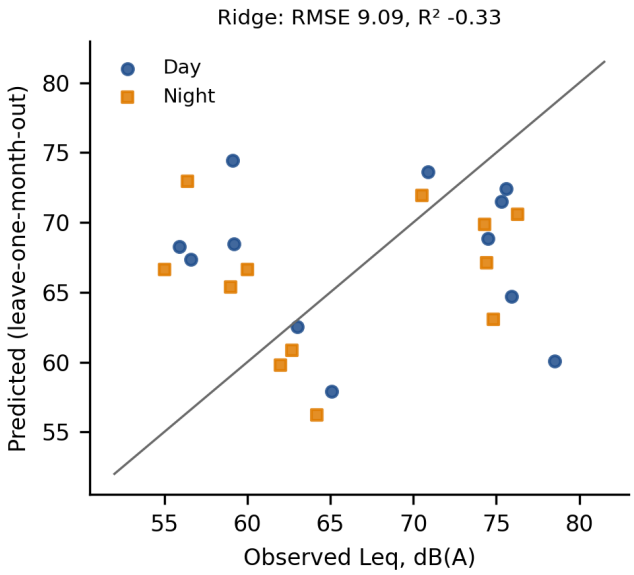


Fig. 4. Observed versus leave-one-month-out predicted Leq for ridge regression; grey line = perfect prediction.

Table 5. OLS coefficients on all 24 observations (R^2 = 0.32, 19 residual d.f.). Indicative only: monthly residuals may be autocorrelated.

Term	Coef. (dB)	95% CI	p
Intercept (day)	+68.20	63.5 to 72.9	<0.001
Night	-1.67	-7.9 to 4.6	0.582
Festival season	-4.40	-15.0 to 6.2	0.394
Month sin	+4.47	-0.5 to 9.5	0.078
Month cos	+3.23	-1.8 to 8.3	0.193

In the descriptive OLS fit (Table 5) every interval for night, festival and the seasonal terms includes zero; the data are consistent with, but do not demonstrate, such effects. The in-sample R^2 of 0.32 is not borne out out-of-sample (Table 4).

6.3 Classifying limit exceedance (Protocol A)

Table 6. Leave-one-month-out classification of limit exceedance (18 of 24 observations exceed). Best F1 in bold.

Model	Acc.	Prec.	Rec.	F1	AUC
Period majority	**0.750**	**0.750**	**1.000**	**0.857**	–
OLS then threshold	0.667	0.750	0.833	0.789	–
Logistic reg.	0.708	0.762	0.889	0.821	0.519
Decision tree	0.625	0.737	0.778	0.757	0.435
k-NN	0.667	0.778	0.778	0.778	0.528
Gauss. NB	0.708	0.789	0.833	0.811	0.444

The majority-by-period rule (day: within limit; night: exceeds) achieved the best F1 (0.857); logistic regression reached F1 0.821 with accuracy 0.708 but an AUC of only 0.52, i.e. almost no ability to rank months. The tree and k-NN were worse (F1 0.757 and 0.778), and thresholding the regression output was worse than classifying directly (F1 0.789). The odds ratio for night in the logistic model is 2.7, but it is estimated from only 24 rows.

6.4 One-month-ahead forecasting (Protocol B)

Table 7. Rolling-origin forecast of L_{eq} (dB(A)), October 2019–March 2020; 12 predictions. Best RMSE in bold.

Model	MAE	RMSE	R^2
Persistence	6.54	7.91	-0.74
Period mean	6.79	7.64	-0.62
OLS	6.13	8.35	-0.93
Ridge	**5.02**	**6.46**	**-0.16**
Decision tree	6.18	7.11	-0.40
k-NN	5.73	6.94	-0.34

Ridge regression had the lowest forecast RMSE (6.46, R^2 -0.16), ahead of the tree (7.11), k-NN (6.94), the period mean (7.64) and persistence (7.91); OLS was worst (8.35). Persistence is poor because consecutive months are weakly related (Spearman ρ = -0.17 between L_{eq} and its previous-month value). With only 12 predictions, however, differences of 1–2 dB are not statistically meaningful, and every R^2 is still negative (Fig. 5).

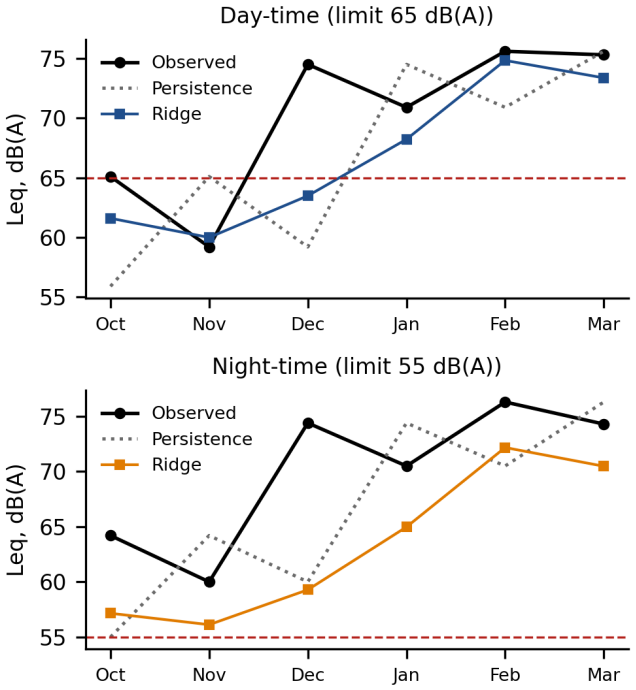


Fig. 5. Rolling-origin forecasts of monthly L_{eq} for day (top) and night (bottom): observed, persistence and ridge. Red dashed: limit.

Table 8. Rolling-origin classification of exceedance (12 predictions; 11 exceedances).

Model	Acc.	Prec.	Rec.	F1
Persistence	0.667	0.889	0.727	0.800
Period majority	**0.750**	**0.900**	**0.818**	**0.857**
OLS then threshold	0.667	1.000	0.636	0.778
Logistic reg.	0.667	0.889	0.727	0.800
Decision tree	0.417	0.833	0.455	0.588
k-NN	0.667	0.889	0.727	0.800
Gauss. NB	0.750	0.900	0.818	0.857

For exceedance, the majority rule is again best (F1 0.857) and no learned classifier improves on it (Table 8). 92% of the forecast observations exceeded the limit against 60% in the early training months, a base-rate shift that favours rules that always predict the common class.

6.5 Sensitivity

Table 9. Sensitivity of the forecasting RMSE to hyper-parameters and to the one row with $L_{max} = L_{eq}$.

Setting	Forecast RMSE (B)
k-NN, k = 2 / 3 / 5	8.54 / 6.94 / 5.53
Tree depth 2 / 3 / 4	6.15 / 7.11 / 6.67
Ridge λ = 0.1 / 1 / 10	8.00 / 6.46 / 5.59
July-2019 day row dropped: OLS / ridge / tree / k-NN	5.91 / 5.16 / 7.85 / 7.44
Persistence (reference)	7.91

The forecasting ranking is unstable: ridge ranges from 5.59 (λ = 10) to 8.00 (λ = 0.1), the latter worse than persistence, and k-NN from 5.53 to 8.54. We therefore do not claim that any learned model forecasts better than the baselines. Only the July-2019 day row could be dropped as a quality test, since the nine rows with L_{min} = 21.6 dB(A) make up three quarters of the day-time data.

7 Discussion

What the data show. The clearest result is regulatory rather than predictive: the 55 dB(A) night limit was exceeded in 11 of 12 months and the 65 dB(A) day limit in 7 of 12. Since night limits protect sleep [2–4], persistent night-time non-compliance in a commercial area is the finding most relevant to a city administration. **What did not work.** With 24 observations and only month, period and festival information, learned models did not beat a period mean or a majority rule in explanation or classification, and their forecasting advantage was not stable. This is a statement about the data size, not about the algorithms.

Limitations. (i) Only twelve monthly values from one station; all intervals are wide and p-values are indicative. (ii) No weekday analysis is possible from monthly aggregates. (iii) Nine day-time rows carry a constant L_{min}, so measurement quality is uncertain. (iv) The festival flag (October–November) is coarse. (v) The data end as the 2020 lockdown began, so its effect cannot be studied. (vi) No traffic, weather or event data were available. Extending

the series with the following year's monthly file would be the most valuable next step.

Ethics and SDG impact. The data are aggregate environmental measurements with no personal information. The main risk is over-interpretation of a very small sample, so we report baselines, intervals and failures. Better use of open noise data supports SDG 11 (sustainable cities, including target 11.6) and SDG 3 (health and well-being).

8 Conclusion and Future Work

Using one open CPCB/KSPCB monthly file for a commercial-area station (April 2019–March 2020), we found that night-time noise exceeded the prescribed limit in 11 of 12 months and day-time noise in 7 of 12, and that classical models (ridge R^2 = -0.33; logistic regression F1 = 0.82, AUC = 0.52) do not outperform trivial baselines on such a short series. Future work should use daily or hourly data (enabling weekday and festival-day analysis), longer records and several stations, add traffic and weather covariates, and verify the instrument anomalies with the monitoring agency.

Declarations

Data availability. The file was obtained from the OGD Platform India resource "Noise levels measured at CAAQMS of CPCB at BWSSB (Commercial Area)" [5]; the cleaned table, predictions and all code (main.py) are in the assignment folder [student to add GitHub link].

Conflict of interest. The author declares no conflict of interest. **Funding.** None.

Use of AI tools. [Student to edit and confirm.] An AI assistant (Claude, Anthropic) was used to write the analysis code, execute the experiments and draft this report from the supplied data and template; every number in the paper is produced by the accompanying code. The author must state precisely what was done by them, verify all content and take full responsibility for it.

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Appendix A. Monthly data used

Table A1. Monthly values (dB(A)); Over = exceeds limit (Y/N) for day / night. Day L_{min} = 21.6 for Apr–Dec 2019 is flagged (Section 3.2).

Month	Day Leq	Lmin	Lmax	Night Leq	Lmin	Lmax	Over (D/N)
Apr-19	59.1	21.6	109.9	56.4	49.9	73.6	N / Y
May-19	75.9	21.6	106.8	74.8	51.4	78.3	Y / Y
Jun-19	56.6	21.6	95.1	59.0	38.2	92.0	N / Y
Jul-19	78.5	21.6	78.5	62.0	55.4	80.3	Y / Y
Aug-19	63.0	21.6	99.0	62.7	39.6	86.5	N / Y
Sep-19	55.9	21.6	89.7	55.0	22.9	91.4	N / N
Oct-19	65.1	21.6	104.4	64.2	49.4	87.7	Y / Y
Nov-19	59.2	21.6	76.8	60.0	48.9	75.0	N / Y
Dec-19	74.5	21.6	89.7	74.4	59.3	87.1	Y / Y
Jan-20	70.9	53.3	90.5	70.5	46.8	93.9	Y / Y
Feb-20	75.6	58.7	92.9	76.3	64.1	87.9	Y / Y
Mar-20	75.3	43.1	100.4	74.3	53.2	98.1	Y / Y